

The Sharp Small-Noise Hilbert–Schmidt Mass of the Folded Gaussian Operator

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Abstract

We compute the exact leading Hilbert–Schmidt mass of the continuum folded Gaussian operator. In the natural $L^2(0, 1)$ geometry, $\sigma \|K_\sigma\|_2^2 \rightarrow (2\sqrt{\pi})^{-1}$; after the Hardy scaling $r_H = 0.85$, the constant is $0.390442618372\dots$. Hence the raw family cannot converge in $\mathcal{S}_2$ at zero noise. The conclusion is scoped: it does not exclude a rank-growing cloud quotient or a cancellation-sensitive trace theorem.

1 Exact row formula

Let $u_c = 1.5436890126920753\dots$, $a = f(x) = 1 - u_c x^2$, and let $k_\sigma(x, y)$ be the normalized folded row

$$k_\sigma(x, y) = \frac{e^{-(y-a)^2/(2\sigma^2)} + e^{-(y+a)^2/(2\sigma^2)}}{Z_\sigma(a)}, \quad 0 \leq y \leq 1.$$

Direct Gaussian integration gives

$$\int_0^1 k_\sigma(x, y)^2 dy = \frac{J_\sigma(a)}{Z_\sigma(a)^2},$$

where

$$J_\sigma(a) = \frac{\sigma\sqrt{\pi}}{2} \left[\operatorname{erf} \frac{1-a}{\sigma} + \operatorname{erf} \frac{1+a}{\sigma} + 2e^{-a^2/\sigma^2} \operatorname{erf} \frac{1}{\sigma} \right],$$

$$Z_\sigma(a) = \sigma\sqrt{\frac{\pi}{2}} \left[\operatorname{erf} \frac{1-a}{\sqrt{2}\sigma} + \operatorname{erf} \frac{1+a}{\sqrt{2}\sigma} \right].$$

2 Sharp law

Theorem 2.1. *For $A_\sigma = r_H^{-1} K_\sigma$,*

$$\lim_{\sigma \downarrow 0} \sigma \|A_\sigma\|_{\mathcal{S}_2}^2 = \frac{1}{2\sqrt{\pi} r_H^2} = 0.3904426183721497\dots$$

In particular no sequence A_{σ_j} with $\sigma_j \downarrow 0$ can converge in $\mathcal{S}_2$.

Proof. For almost every x , $a \in (-1, 1) \setminus \{0\}$. In the row formula the opposite folded Gaussian and both boundaries become exponentially negligible, so $\sigma J_\sigma(a)/Z_\sigma(a)^2 \rightarrow (2\sqrt{\pi})^{-1}$. The normalizer is uniformly bounded below by a half-Gaussian mass, while $(p+q)^2 \leq 2(p^2+q^2)$ gives $\sigma \int k_\sigma(x, y)^2 dy \leq C$ uniformly in x and small σ . Dominated convergence over $x \in [0, 1]$ proves the limit. Hardy scaling multiplies the square norm by r_H^{-2} . $\square$

An independent interior-window estimate gives $\|K_\sigma\|_2^2 \geq 0.0700325262/\sigma$, while the inherited upper bound is $4.5135/\sigma$. The exact limit identifies the sharp leading constant.

3 Audit and boundary

One-dimensional quadrature gives ratios to the limit decreasing from 1.1579 at $\sigma = 0.04$ to 1.0104 at $\sigma = 0.00025$. This is a reproduction check, not part of the proof. The theorem closes only the raw zero-noise $\mathcal{S}_2$ option; Gates A–E remain open.